

Unlocking Revenue: Analyzing Hotel Booking Cancellations

This project analyzes 119,390 hospitality booking records to identify key factors influencing cancellations and provide data-driven recommendations to reduce revenue loss.

Project Overview & Dataset Summary

Dataset Size

119,390 Rows

Data Points

31 Columns

Key Focus

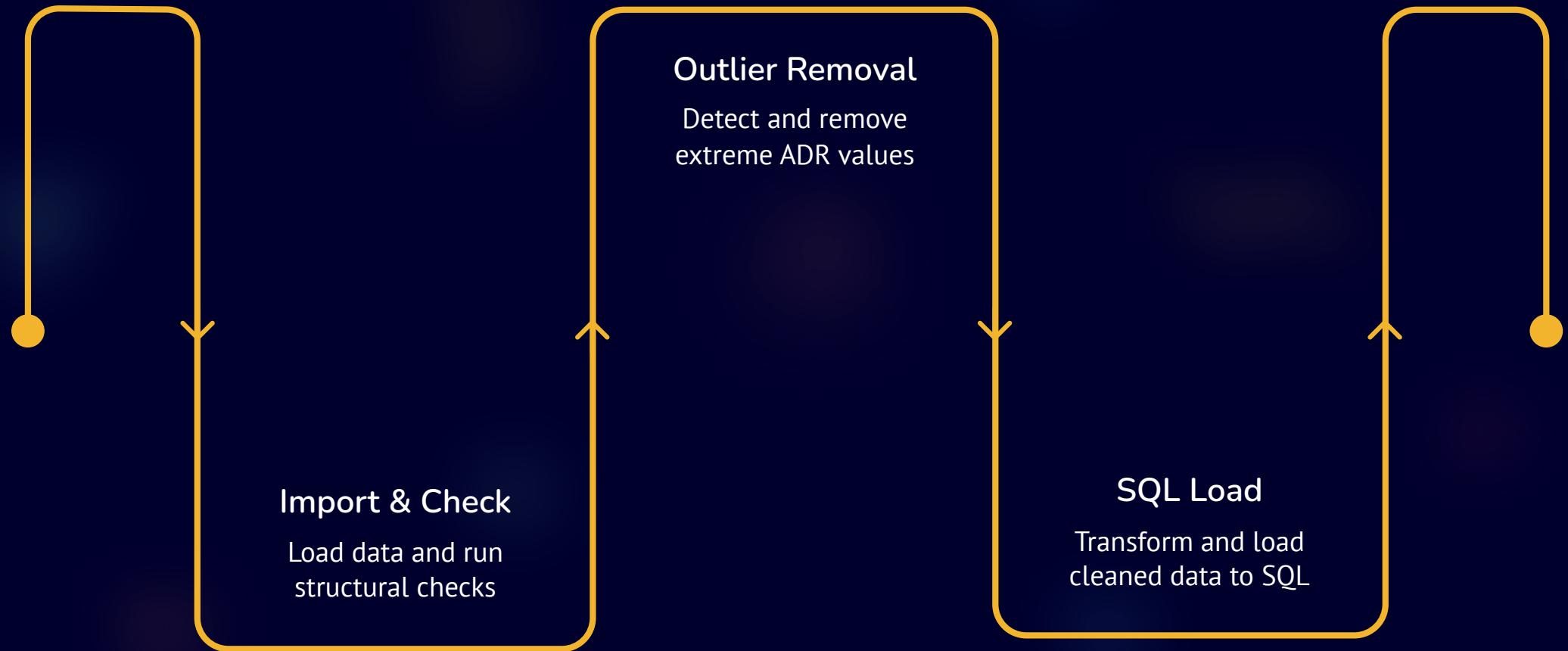
Cancellation Factors

Our analysis delves into a comprehensive dataset with 119,390 records and 31 features, spanning multiple countries. The core objective is to pinpoint the drivers behind hotel booking cancellations.

Key feature groups include booking timing, customer history, pricing strategies, distribution channels, and specific hotel characteristics. Understanding these elements is crucial for developing effective mitigation strategies.



Data Preparation & Cleaning



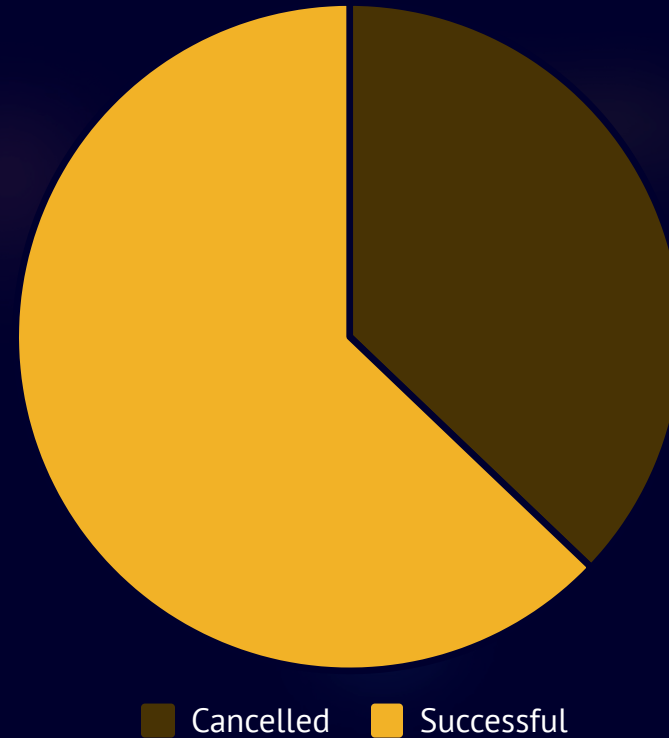
Our data preparation involved importing data with Pandas, performing structural and statistical checks, and converting date formats. We detected and removed extreme outliers in Average Daily Rate (ADR) to ensure data integrity. Finally, the cleaned dataset was loaded into an SQL database for robust analytical queries and backup.

Exploratory Data Analysis: Key Insights

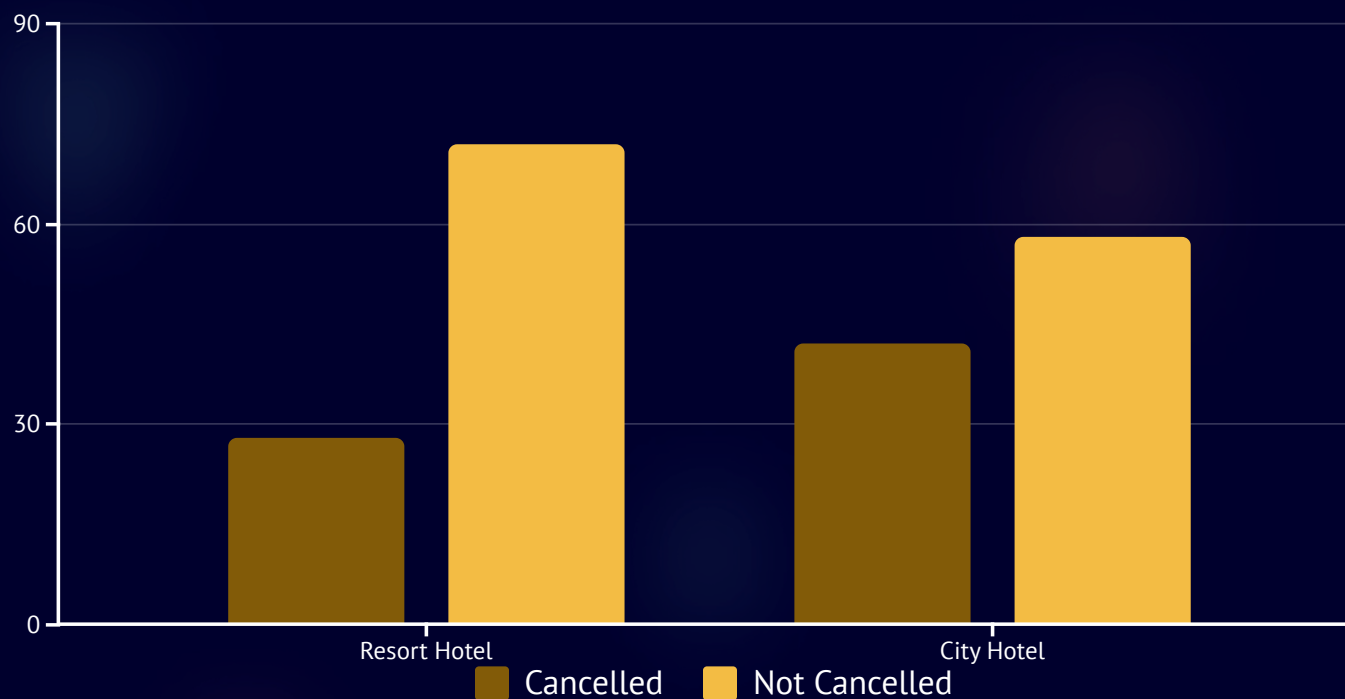
Overall Cancellation Rate

Our analysis reveals a significant cancellation rate, indicating a substantial revenue leakage risk for hotels.

- Cancelled bookings: 37.13%
- Successful bookings: 62.86%
- Includes 1,203 no-shows and 42,949 cancellations.



Hotel Type vs. Cancellation



City Hotels exhibit a significantly higher cancellation rate compared to Resort Hotels. This disparity suggests potential issues such as pricing sensitivity, overbooking practices, or unmet customer expectations in urban settings.

Addressing these specific challenges in City Hotels could lead to a substantial reduction in overall cancellation rates.



Revenue & Pricing Trends

Dynamic pricing, while effective for revenue generation, can increase booking volatility if not carefully managed. Extreme fluctuations in ADR, especially during peak seasons, may contribute to higher cancellation rates.

1

City Hotel Revenue Peak

City hotel revenue peaked in 2016 with a +28.51% increase.

2

Increased Sellable Rooms

Average sellable rooms consistently increased year-over-year.

3

ADR Spikes

ADR spikes align with weekends and peak seasons, indicating dynamic pricing strategies.

4

Highest ADR in August

August consistently recorded the highest ADR, averaging around \$111.

Country-wise Cancellations: Portugal Focus



Our analysis highlights a significant geographical concentration of cancellations. Portugal alone accounts for approximately **69.86%** of total cancellations, representing 26,751 bookings.

This regional trend is likely driven by factors such as flexible cancellation policies, high usage of Online Travel Agencies (OTAs), and a customer base that is leisure-driven and price-sensitive.

Targeted strategies for this region are essential to mitigate revenue loss.

Recommendations to Reduce Cancellations



Smart Cancellation Policies

Introduce tiered rules: free cancellation for shorter windows, partial penalties closer to arrival. Reduce flexibility for high-risk segments.



Lead-Time Risk Control

Identify long lead-time bookings as high-risk. Apply small advance deposits, send reminder emails, and offer incentives for early confirmation.



Target City Hotel Cancellations

Improve room description accuracy, use overbooking prediction models, and offer non-refundable discounted plans for price-sensitive guests.



Geo-Specific Strategy (Portugal)

Tighten cancellation rules for high-cancellation countries. Partner with local OTAs for prepaid bookings and offer localized loyalty discounts.

Further Recommendations for Commitment



Pricing Stability

Avoid extreme ADR fluctuations during peak seasons. Introduce price-lock options for early bookings.



Leverage Customer History

Reward repeat guests with flexible benefits. Restrict flexibility for customers with previous cancellations.



Value Add-ons

Promote free parking, meal upgrades, or special request prioritization to increase perceived value and commitment.

Business Impact & Conclusion

Implementing these data-driven recommendations can significantly benefit hotels by:

- Reducing cancellation rates by 10–20%.
- Improving occupancy predictability for better resource allocation.
- Increasing realized revenue without additional marketing spend.

By understanding and addressing the root causes of cancellations, hotels can optimize operations and enhance profitability.

